Т13: График мощности критерия

Используем

$$\tilde{F} \equiv F(138, 999), \quad \gamma = \frac{\sigma^2_{\text{er}}}{\sigma^2_{\text{ев}}},$$

$$W(\gamma) = \tilde{F}\left(\frac{\tilde{F}_{\alpha/2}}{\gamma}\right) + 1 - \tilde{F}\left(\frac{\tilde{F}_{1-\alpha/2}}{\gamma}\right).$$

In [3]:
```python
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as st

alpha = 0.05
dfn, dfd = 138, 999

F_alpha_2 = st.f.ppf(alpha / 2, dfn, dfd)
F_1_minus_alpha_2 = st.f.ppf(1 - alpha / 2, dfn, dfd)

def W(gamma):
    return st.f.cdf(F_alpha_2 / gamma, dfn, dfd) + 1 - st.f.cdf(F_1_minus_alpha_2 / gamma,
dfn, dfd)

gamma_grid = np.linspace(0.2, 3.0, 2000)
power = W(gamma_grid)

plt.figure(figsize=(10, 5.5))
plt.plot(gamma_grid, power, color='royalblue', linewidth=2.2, label='W(γ)')
plt.axvline(1.0, color='black', linestyle='--', linewidth=1.2, label='γ = 1')
plt.axhline(alpha, color='firebrick', linestyle=':', linewidth=1.4, label='α = 0.05')
plt.xlim(gamma_grid[0], gamma_grid[-1])
plt.ylim(0, 1.02)
plt.xlabel('γ')
plt.ylabel('W(γ)')
plt.title('Функция мощности')
plt.grid(alpha=0.25)
plt.legend()
plt.show()
```

